

Key Factors Influencing Student Exam Scores

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Math564: Regression

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Abstract/Introduction

This report analyzes factors that influence student exam scores using multiple regression analysis. The main objective of this project is to determine the impact of Hours_Studied, Attendance, and Peer_Influence (categorized as positive, neutral, and negative) on exam performance. We first fit a regression model to understand the significance of these variables and assess how well they explain variations in exam scores. We found that Hours_Studied and Attendance seems to be strong positive predictors, while Neutral and Negative Peer_Influence were found to negatively affect exam scores.

The initial model showed some issues with heteroscedasticity. To address this, we applied a log transformation to the response variable (Exam_Score), leading to an improved model with better fit and more consistent error variance. The transformed model explained approximately 59.3% of the variation in exam scores, compared to 55.1% of the variance explained by the initial model.

Overall, the analysis provides valuable insights into academic performance and suggests where intervention, such as promoting study habits and positive peer influences, could be effective in improving students' performance in exams.

The original dataset contains various factors that contribute to student performance in exams. It includes information on study habits, attendance, family income, and etc.

We will be looking at the following key attributes:

- Exam_Score : The dependent variable measuring students' performance in the final exam.
- Hours_Studied: A continuous variable representing the number of hours a student spends on studying per week.
- Attendance: A continuous variable capturing a student's attendance in terms of percentage.
- Peer_Influence: A categorical variable with three levels (Positive, Neutral, Negative) indicating the influence of peers on academic performance.

$$\text{Exam Score} = \hat{\beta}_0 + \hat{\beta}_1 \text{Hours Studied} + \hat{\beta}_2 \text{Attendance} + \hat{\beta}_3 \text{Peer Influence}_1 + \hat{\beta}_4 \text{Peer Influence}_2$$

Dummy variables:

Peer Influence_1		Peer Influence_2	
1	If Peer Influence is Neutral	1	If Peer Influence is Negative
0	Otherwise	0	Otherwise

Initial Analysis: Model Fitting and Interpretation

```
##
## Call:
## lm(formula = Exam_Score ~ Hours_Studied + Attendance + Peer_Influence,
##     data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -6.465 -1.289 -0.171  0.984 32.146
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   46.014370    0.252222  182.436 < 2e-16 ***
## Hours_Studied    0.292431    0.005354   54.614 < 2e-16 ***
## Attendance      0.197527    0.002779   71.086 < 2e-16 ***
## Peer_InfluenceNeutral -0.516871    0.072124  -7.166 8.53e-13 ***
## Peer_InfluenceNegative -1.031788    0.086668 -11.905 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.607 on 6602 degrees of freedom
## Multiple R-squared:  0.5513, Adjusted R-squared:  0.551
## F-statistic: 2028 on 4 and 6602 DF, p-value: < 2.2e-16
```

Based on the summary for the initial model, we can conclude that the variables Hours_Studied, Attendance, and Peer_Influence are all significant predictors of Exam_Score, all of them have p-values less than 0.001. Since all predictors contribute significantly to the model, we can interpret the value of $\hat{\beta}_i$ for $i = 0, 1, 2, 3, 4$ as the following:

- Intercept ($\hat{\beta}_0 = 46.014370$)
When Hours_Studied and Attendance are both zero, and Peer Influence is positive the expected Exam_Score is approximately 46.
- Hours_Studied ($\hat{\beta}_1 = 0.292431$)
For each additional hour a student spends on studying each week, the Exam_Score is expected to increase by 0.29 points, assuming Attendance and Peer Influence remain constant. This positive coefficient indicates that more time spent on studying leads to higher exam scores.
- Attendance ($\hat{\beta}_2 = 0.197527$)
For each percentage point increase in Attendance, the Exam_Score is expected to rise by 0.2 points, holding Hours_Studied and Peer Influence constant. This suggests that higher attendance has a positive impact on exam performance.
- Peer_Influence Neutral ($\hat{\beta}_3 = -0.516871$)
Students with a level of neutral peer influence score, on average, 0.52 points lower than students with a level of positive peer influence, holding Hours_Studied and Attendance constant. This indicates that a neutral peer influence negatively impacts exam scores compared to positive peer influence.
- Peer_Influence Negative ($\hat{\beta}_4 = -1.031788$)
Students with a level of negative peer influence score, on average, 1.03 points lower than students with a level of positive peer influence, holding Hours_Studied and Attendance constant. This indicates that a negative peer influence significantly lowers exam

performance compared to a positive peer influence.

Goodness-of-Fit Metrics:

- R^2 : 0.5513
The R^2 of the initial model indicates that approximately 55.13% of the variation in Exam_Score is explained by our model using Hours_Studied, Attendance, and Peer Influence. The model does a decent job of capturing what affects exam scores, but it's not perfect. There are still other factors out there that influence exam performance, which our model did not account for.
- Adjusted R^2 : 0.551
The adjusted R^2 value is almost the same as the regular R^2 value. It accounts for the number of predictors in the model and helps us know if adding more variables would be useful. Since it is very close to the regular R^2 value, this suggests that all the predictors we included are meaningful and not just adding noise.
- F-statistic: 2028 with p-value $< 2.2e-16$
The F-statistic test tells us whether our model is useful overall. Since we have a very large F-value with an extremely small p-value, this tells us that our model as a whole is statistically significant. In other words, the combination of Hours_Studied, Attendance, and Peer_Influence does a great job at predicting exam scores and it's not just due to random chance.

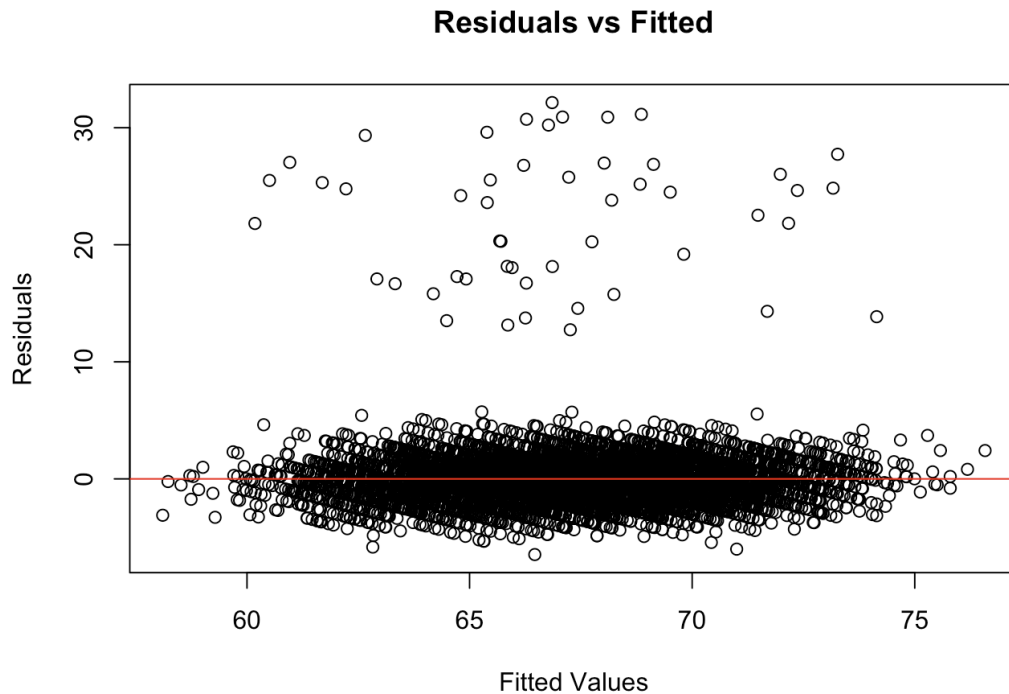
Overall, the model is fairly good at explaining Exam_Score, but it's not capturing everything. The results show that our predictors are important, but there is still room to improve the model by considering other influences on exam performance.

Regression Diagnostics: Model Assumptions and Issues

```
##
## Durbin-Watson test
##
## data: model1
## DW = 2.0033, p-value = 0.5527
## alternative hypothesis: true autocorrelation is greater than 0
```

1. Autocorrelation

I used the Durbin-Watson test to assess whether the residuals of the model were autocorrelated. Autocorrelation refers to the situation where errors are not independent of each other, which can be problematic. The DW value is close to 2 with a high p-value, this indicates that there is no significant autocorrelation in the residuals. In other words, our errors do not have a systematic pattern and are fairly random, which meets one of the key regression assumptions.



2. Heteroscedasticity

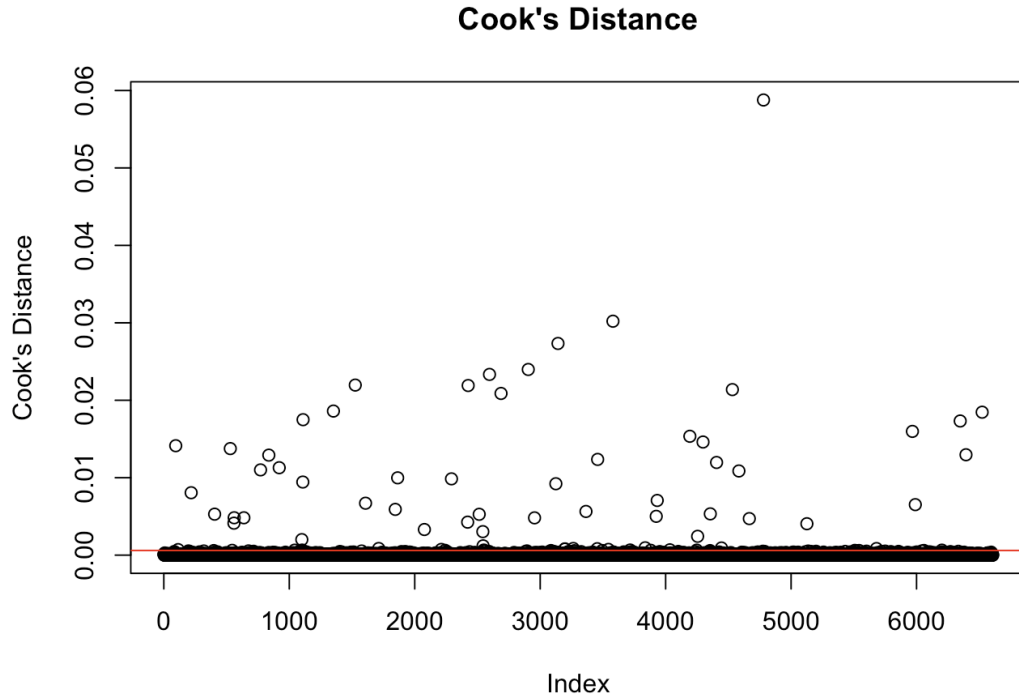
I checked for heteroscedasticity using the residuals vs. fitted plot. In a well behaved model, the residuals should be evenly spread around zero across all levels of fitted values. Looking at the residuals vs. fitted plot for the initial model, we can see that the residual values range from about -5 to 32. This wide range of residuals suggests that the model does not perform very well, and there seems to be influential points affecting the overall model fit. While the plot does not show a funnel-like pattern indicating heteroscedasticity, the cluster of high value residuals suggest that there may be observations that have much larger errors compared to others.

##		GVIF	Df	$GVIF^{(1/(2*Df))}$
##	Hours_Studied	1.000232	1	1.000116
##	Attendance	1.000888	1	1.000444
##	Peer_Influence	1.000928	2	1.000232

3. Multicollinearity

I used Variance Inflation Factor (VIF) values to detect multicollinearity among predictors in the initial model. Multicollinearity occurs when predictor variables are highly correlated with each other, this would make it difficult for us to access the individual effect of each predictor. In general, VIF values greater than 5 or 10 indicate problematic amount of collinearity, and VIF values close to 1 indicate no multicollinearity. In our case, all VIF values are very close to 1, suggesting that there is no issue with multicollinearity in the initial model. In other words, the predictors (Hours_Studied, Attendance and Peer_Influence) are not highly correlated with one another, and each

variable provides unique information to the model. Since multicollinearity is not a problem, the standard errors of the regression coefficients are reliable, and the estimates for each predictor can be interpreted without concern for inflated variability.



4. Influential Points

I used Cook's Distance to determine influential points in the initial model. Cook's Distance measures how much the regression coefficients change if a specific data point is removed, this helps to identify influential data points that have a significant impact on the model's estimates. Points with a Cook's Distance greater than 1 or $\frac{4}{n}$ (n is the number of observations) are considered as highly influential. Even though we don't have data points having a Cook's Distance of 1 or higher, we can see from the plot that we have a few points that lies above the red line (indicates the value of $\frac{4}{n}$). Our plot does not indicate a clear implication that there are influential points in our initial model, but we can still try to refine our model to make it more stable and reliable.

Remediation: Addressing Model Issues

Our initial model did a decent job in explaining the students' performance in the final exam, but the diagnostic test showed that there may be some potential problems. The Residuals vs. Fitted plot showed a wide spread of errors, suggesting that there may be a problem with heteroscedasticity.

To fix this, I applied a logarithm transformation to the response variable, Exam_Score. This method helps stabilize the variance, reduce the effect of outliers, and make the relationship

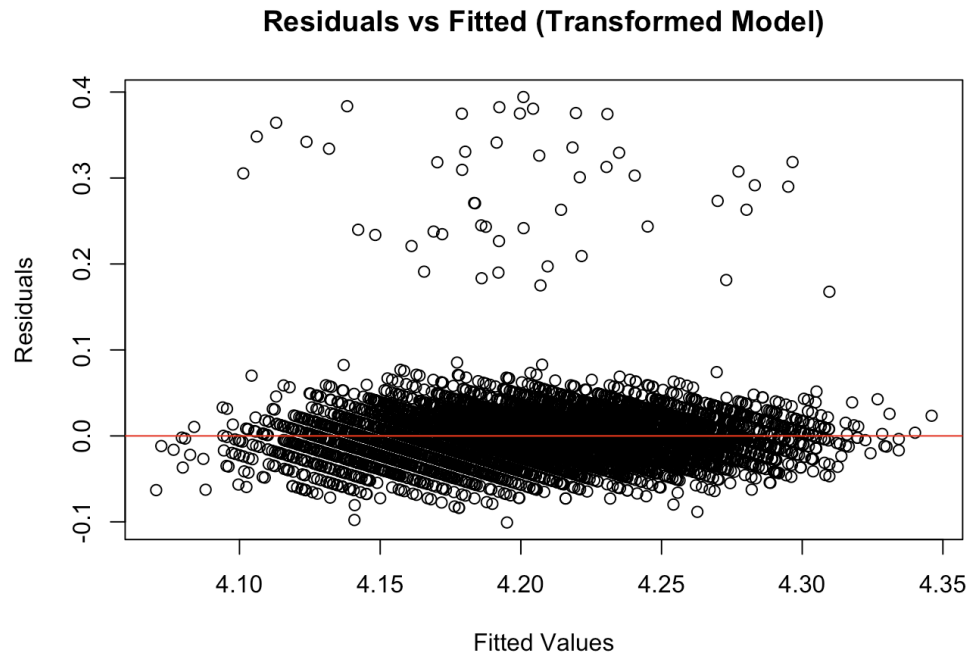
between variables clearer. By transforming the data, we can create a more reliable model with better performance and fewer problems.

```
##
## Call:
## lm(formula = ExamScore_log ~ Hours_Studied + Attendance + Peer_Influence,
##     data = df)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.10073 -0.01866 -0.00186  0.01517  0.39422
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    3.890e+00  3.457e-03 1125.258 < 2e-16 ***
## Hours_Studied    4.365e-03  7.339e-05  59.474 < 2e-16 ***
## Attendance       2.946e-03  3.808e-05  77.365 < 2e-16 ***
## Peer_InfluenceNeutral -7.629e-03  9.885e-04  -7.717 1.36e-14 ***
## Peer_InfluenceNegative -1.535e-02  1.188e-03 -12.921 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.03573 on 6602 degrees of freedom
## Multiple R-squared:  0.5928, Adjusted R-squared:  0.5926
## F-statistic: 2403 on 4 and 6602 DF,  p-value: < 2.2e-16
```

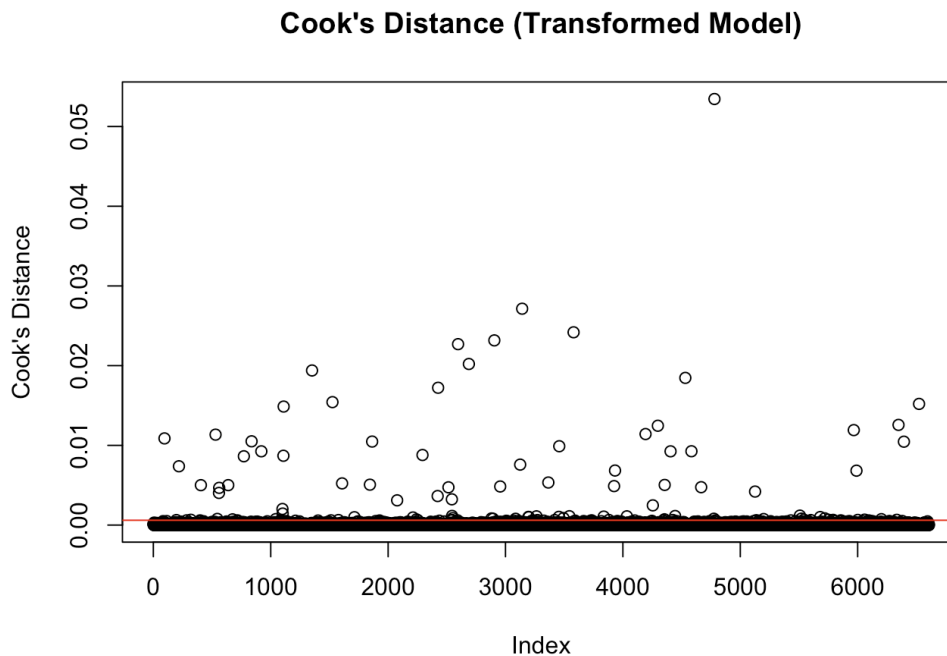
This picture shows the output of the summary for the transformed model.

From the output of the transformed model, we can observe that the p-value for Neutral Peer Influence decreased from $8.53\text{e-}13$ in the initial model to $1.36\text{e-}14$ in the transformed model. This decrease in the p-value suggests that the effect of Neutral Peer Influence on exam scores is even more statistically significant in the transformed model. In other words, the log transformation improved the clarity of this predictor's impact, making it more certain that Neutral Peer Influence has a meaningful and negative effect on exam scores. Additionally, the p-values for the other variables (Hours Studied, Attendance, and Negative Peer Influence) remained at $<2\text{e-}16$, which is extremely significant. This shows that the transformation did not weaken the statistical strength of these variables. Instead, it confirms that their effects on exam scores are consistently strong and reliable across both models.

Additionally, we can observe that the R^2 value has increased from 0.5513 to 0.5928, and the adjusted R^2 value has increased from 0.551 to 0.5926. This suggests that the transformed model explains a higher proportion of the variance in Exam_Score, the log transformation improved the model's fit and helped capture more of the variation in Exam_Score. We also see that the Residual Standard Error (RSE) has decreased from 2.607 to 0.03573. The RSE is much lower in the transformed model, suggesting a better fit and less variability in the residuals. Finally, we can see that the F-statistic value has increased from 2028 to 2403. This indicates that the log transformation has enhanced the model's explanatory power. With the p-values remaining $<2.2\text{e-}16$ in both models, we can conclude that the model remains statistically reliable and relevant, despite the transformation.



The Residuals vs. Fitted plot for the transformed model shows a more consistent spread of residuals, ranging from -0.1 to 0.4 on the y-axis. The range of the residuals is much smaller in the transformed model compared to the initial model, which ranges from -5 to 32. This indicates that the log transformation effectively stabilized the variance. Overall, the log transformation has improved the residual distribution, reducing heteroscedasticity and making the residuals more consistent. This leads to a more reliable regression model.



From the Cook's Distance plot for the transformed model, we have fewer points with Cook's Distance values above 0.02. The range of influential points has become narrower, indicating that the model is now less affected by individual observations. Comparing the two models, the transformed model shows a better distribution with fewer extreme values. Therefore, we can conclude that the transformed model is more reliable, as the influence of outliers has been reduced. This suggests that the log transformation was effective in improving the model's performance.

```
##
## Durbin-Watson test
##
## data: model_transformed
## DW = 2.0008, p-value = 0.5132
## alternative hypothesis: true autocorrelation is greater than 0
```

Comparing the Durbin-Watson test results for the two models, both models have DW statistics very close to 2, indicating that there is no significant autocorrelation in the residuals. The high p-values further confirm that the errors are independent in both cases.

```
##              GVIF Df GVIF^(1/(2*Df))
## Hours_Studied 1.000232 1      1.000116
## Attendance    1.000888 1      1.000444
## Peer_Influence 1.000928 2      1.000232
```

The VIF values remain the same for both models, all very close to 1. This indicates that multicollinearity is not an issue, and the predictor variables are not highly correlated. Since the VIF values remain the same for both models, we can conclude that the log transformation did not affect the relationships among the predictors.

Summary and Conclusions

The analysis of student exam performance using regression modeling revealed several important insights, and we managed to improve the model through a logarithmic transformation of the response variable. Initially, we found that the number of hours students spend studying per week and their attendance are highly significant predictors, increased study time and better attendance are tied to higher exam scores. Additionally, the level of peer influence also matters. Students who had neutral or negative peer influences tended to do worse compared to those with positive peer influence. These findings stayed consistent, even after we transformed the model.

The biggest improvement came from using a log transformation on Exam_Score. This made the model fit the data better. We observed that the adjusted R^2 value went up from 0.551 to 0.5926, which means the model explained more of why scores vary. The F-statistic also jumped from 2028 to 2403, proving that the model got stronger and could better explain the relationships between study habits, attendance, and peer influence.

In the initial model, the range of the residuals was pretty wide, but we managed to shrink the range of the residuals in the transformed model. This indicates that the errors have become more stable and our model's predictions are more dependable.

When we checked for influential points using Cook's Distance, we saw improvements too. The transformation made the model less sensitive to outliers and data points that could have an impact on the results. This made the model more solid and less likely to be thrown off by unusual scores.

The Durbin-Watson test showed that there was no problem with autocorrelation. The values stayed close to 2 with high p-values for both models, meaning that our errors were independent. And we checked for multicollinearity with VIF, all the values were near 1 for both models, which means there wasn't a problem with our predictors being too closely related to each other. Overall, the log transformation significantly improved the regression model's performance, making it a more accurate and reliable tool for understanding the factors that impact student exam scores. The transformed model is well-behaved, with reduced heteroscedasticity, fewer influential points, and stable predictor significance. The refined model provides a clearer understanding of how study habits, attendance and peer influence affect academic success. To improve this project, we could look into adding more meaningful predictors to boost the model's performance even further. Although we have a decent adjusted R^2 value of 0.5926 for our transformed model, meaning that the model explains about 59.3% of variation in exam scores, there is still room to make it better. We can include additional variables that might influence student performance, such as time spent on extracurricular activities, parental involvement, or access to study resources, could help increase the R^2 value. Adding these extra predictors might capture more of the factors affecting exam scores, giving us a fuller and more accurate understanding of what drives student success.

References

"Student Performance Factors." *Kaggle*, 4 Nov. 2024,
www.kaggle.com/datasets/languyn123/student-performance-factors